

Chapter 4

Budgeting in the United States

In this chapter, we cover budgeting specific to the U.S. context. We introduce the budget function classification followed by various aspects of the federal budget such as process, key institutions, evolution of historic and current budget laws, budget cycle, and types of budget authority. We conclude this chapter with the introduction of the main social insurance programs and the federal debt.

4.1 Budget Function Classification

The federal budget is categorized around functions, which have the following purpose (GAO, 1998):

... categorize federal spending according to the purpose or mission of government...

Budget functions are not administrative departments since one function can be associated with multiple departments such as national defense is in the Department of War and the Department of Energy (among others). It is also possible to have one department and multiple functions. For example, the Department of War is responsible for national security and natural resources and environment given the large area of land occupied by military bases. The structure of the budget resolution is around 20 functions of which 17 are labeled as broad areas of national need and three are labeled as miscellaneous (e.g., interest payments on federal debt). Examples:

- **Community and Regional Development:** (1) Community development, (2) area and regional development, and (3) disaster relief and insurance
- **Energy:** (1) Energy supply, (2) energy conservation, (3) emergency energy preparedness, and (4) energy information, policy, and regulation

- **General Science, Space and Technology:** (1) General science and basic research and (2) space flight, research, and supporting activities

4.2 Key Institutions in the Federal Budget Process

There are many institutions that are involved in the federal budget process. The main executive branch entity is the Office of Management and Budget (OMB), which was mentioned previously as a major data source. Legislative branch offices, i.e., those reporting to Congress, are the Congressional Budget Office (CBO), Government Accountability Office (GAO), and Congressional Research Service (CRS).

The Office of Management and Budget (OMB) is responsible for the construction of the President's budget. It is (1) setting the funding priorities, (2) evaluates the effectiveness of agency programs, policies, and procedures, and (3) assess competing funding demands among agencies. The OMB submits the President's annual budget and supporting materials to Congress. It supervises and coordinates the administration's procurement, financial management, information, and regulatory policies. The OMB is also tasked with improving the administrative management and developing performance measures. And lastly, it ensures consistency of agency reports, rules, testimony, and proposed legislation with the President's budget and administration policies and priorities.

The Congressional Budget Office (CBO) provides nonpartisan budgetary information and analysis to Congress and its committees. Its responsibilities include preparing an annual five-year economic forecast, reviewing the President's annual budget submission, developing budget baseline projections, preparing financial reports, and evaluating spending and revenue options for reducing the federal budget deficit.

The Government Accountability Office (GAO) is an independent, nonpartisan agency in the legislative branch that assists Congress in overseeing federal programs and the use of public funds. Often described as the congressional watchdog, GAO conducts financial audits, program evaluations, investigations, legal analyses, and other reviews designed to assess the economy, efficiency, effectiveness, and accountability of federal government activities. Much of its work is undertaken at the request of congressional committees or subcommittees or as required by law, and its reports frequently include recommendations for improving federal programs and increasing efficiency. The Congressional Research Service (CRS) is a legislative branch agency within the Library of Congress that provides research and policy analysis directly to members and committees of Congress. CRS supports Congress throughout the legislative process by analyzing existing policies, proposed legislation, legal issues, budget and appropriations matters, and alternative policy options. Its work is intended to be objective and nonpartisan and includes publicly available reports as well as confidential

memoranda, briefings, consultations, and responses to individual congressional inquiries. Unlike GAO, which focuses heavily on oversight and evaluation of government programs and spending, CRS primarily provides research and analysis to help Congress develop, evaluate, and understand legislation and public policy.

4.3 Key Federal Budget Laws

There are some key laws associated with U.S. federal budgeting. Before we present those key laws, let us cover some of the terminology used.

- **Impoundment** refers to the decision of the President to withdraw funds.
- **Rescission** is the proposal by the President to withdraw funds. It requires a majority vote of approval by Congress within 45 days.
- **Deferrals** are temporary delays in the release of funds for obligation. The maximum delay is until the end of the fiscal year. Like the rescission, a deferral requires approval by Congress as well.

More information on rescission and deferrals can be found in [Judicial Review of Rescissions and Deferrals Under the Impoundment Control Act](#).

Let us start with the overarching law governing expenses at the federal level. Article 1 (Section 9) of the U.S. Constitution states the following:

No Money shall be drawn from the Treasury, but in consequence of appropriations made by law; and a regular statement and account of the receipts and expenditures of all public money shall be published from time to time.

Over time, many more laws were enacted with regard to budgets, expenditures, and taxation. The Anti-Deficiency Act (earliest version 1870) prohibits departments to spend more than appropriated by Congress. The [Budget and Accounting Act of 1921](#) created the Bureau of Budget (which is now called OMB) and the GAO. It set the start of the fiscal year to July 1 (which is now October 1). In addition, it established a national budget system and independent audits of government accounts.

The [Congressional Budget and Impoundment Control Act of 1974](#) established budget committees in both the House and Senate, as well as created the Congressional Budget Office (CBO). These committees and the CBO were tasked with enhancing the legislative branch's ability to analyze, oversee, and manage federal fiscal policies. The Act required the President's budget to include a functional classification, which we have mentioned previously, to organize federal spending by purpose (e.g., defense, education, healthcare). The Act also replaced the practice of impoundment, i.e., a situation in which the President could withhold spending of funds appropriated by Congress, with rescission and deferral. Finally, the Act shifted the start of the fiscal year from July 1 to October 1.

The Gramm-Rudman-Hollings Act (also known as the [Balanced Budget and Emergency Deficit Control Act of 1985](#)) established deficit targets to balance the federal budget by 1991. The mechanism to meet the target was called sequestration (“cancellation of budgetary resources”). It triggers automatic spending cuts if total discretionary appropriations exceed budget spending thresholds. It affects all departments and programs by an equal percentage. The original version was ruled unconstitutional by the Supreme Court because it violates *the Constitution’s command that Congress play no direct role in the execution of the laws*.

The Budget Enforcement Act (BEA) was part of the [Omnibus Budget Reconciliation Act of 1990](#). It is a replacement of the Balanced Budget and Emergency Deficit Control Act of 1987 (previous version from 1985) and establishes upper limits on discretionary spending. The Act expired in 2002. The BEA institutionalized so-called pay-as-you-go (PAYGO) rules as a way to manage the federal deficit by requiring offsets for any new spending or tax cuts, making it a central part of fiscal discipline in the 1990s. PAYGO requires that any new legislation affecting mandatory spending or tax revenue must not increase the federal deficit. Specifically, if Congress enacts a law that either increases entitlement spending or decreases revenues (such as through tax cuts), the cost must be offset by either cuts to other mandatory spending programs or by increases in revenue (e.g., through tax hikes). This means that any new spending initiatives or tax reductions must be “paid for” to ensure fiscal discipline and prevent adding to the national debt. Put differently, it required policy changes to be deficit-neutral or deficit-reducing. Although the PAYGO provisions of the Budget Enforcement Act of 1990 expired in 2002, Congress reinstated a similar statutory framework through the Statutory Pay-As-You-Go Act of 2010. The Statutory Pay-As-You-Go Act of 2010 (PAYGO) established budget enforcement procedures intended to prevent new legislation affecting mandatory spending or revenues from increasing the federal deficit. In general, increases in mandatory spending or reductions in revenues must be offset by corresponding spending reductions or revenue increases; otherwise, sequestration can be triggered.

The Budget Control Act of 2011 imposed statutory limits on discretionary spending for fiscal years 2012 through 2021 and established automatic spending reductions, commonly referred to as sequestration, if specified budgetary targets were not met. The legislation also provided for congressional consideration of a constitutional balanced-budget amendment. The Bipartisan Budget Acts of 2013 and 2015 subsequently relaxed some of the discretionary spending constraints established under the Budget Control Act. The 2013 legislation raised discretionary spending limits for fiscal years 2014 and 2015, while the 2015 legislation established higher limits for fiscal years 2016 and 2017, effectively easing the restrictiveness of the original spending caps.

4.4 Federal Budget Cycle

The federal budget cycle involves four steps: (1) Executive preparation, (2) legislative review and approval, (3) budget implementation, and (4) audit and evaluation. This is a structured timetable beginning with the submission of the President's budget no later than the first Monday in February. Six weeks later, the Congressional budget committees report their budget estimates.

Key dates in the process include April 15, by which Congress must complete its budget resolution, and May 15, when the House begins considering appropriations bills. By June 15, reconciliation should be completed, and the House is expected to finalize appropriations by June 30. On July 15, the President submits a mid-session budget review, and the fiscal year officially starts on October 1.

4.4.1 President's Budget Request

The President's budget request is developed by the Office of Management and Budget (OMB) based on submissions from federal agencies. It begins with an outline of the President's overall fiscal policies, detailing expected spending, tax revenues, and projections of deficits or surplus spending.

The budget also provides an outline of the President's priorities for federal programs over the next five years, highlighting areas such as defense, agriculture, and education. Additionally, it includes recommendations for changes in spending and tax policies to align with the administration's goals. More information can be found in the video titled [U.S. Federal Budget Process 101](#) and produced by the American Association for the Advancement of Science (AAAS) as well as the CRS report [The Executive Budget Process Timetable: In Brief](#).

4.4.2 Congressional Budget Resolution

The CBO explains the Congressional Budget Resolution as follows:

... a blueprint to guide Congressional action on budget-related legislation over the course of the year. It does not provide funding for federal programs or change tax law; rather, it sets overall spending and revenue targets, sometimes for as many as 10 years. A budget resolution is not a law, because it is not signed by the President; it takes effect if approved by both Houses of Congress.

The Congressional Budget Resolutions set spending ceilings for 20 budget functions and include revenue estimates. The budget resolution is drafted by the [House](#) and [Senate](#) budget committees, followed by floor votes where amendments are passed by majority vote. Any differences between the House and Senate are resolved in conference committees, and the resolution cannot be filibustered.

The resolution is expected to pass by April 1, but if delayed or not passed, the previous year's resolution remains in effect. Budget resolutions are enforced by allowing House and Senate members to block legislation that violates the

resolution, although this can be waived by a simple majority in the House or 60 votes in the Senate.

Budget reconciliation is used to enforce spending cuts or tax increases specified in the resolution, with reconciliation directives providing detailed instructions for achieving these fiscal targets by a set deadline.

4.4.3 Appropriation Bills

Annual appropriations are necessary for discretionary (but not mandatory) funding. Those appropriation bills become law with the President's signature. The appropriation bills are drafted by the appropriation committees (i.e., [Senate](#) and [House](#)). In the past, the on-time passage of all appropriation bills before the start of the Federal fiscal year has been difficult (see [Congress has long struggled to pass spending bills on time](#) from the Pew Research Center). Since 1977 (the first year of the current system), only four years (1977, 1989, 1995, and 1997) had all appropriation bills passed on time. Often, so-called omnibus bills are passed meaning all appropriation bills are packed into one bill. The [Appropriations Status Table](#) shows the progress of the current bills. [S.3109 - Department of Homeland Security Appropriations Act, 2019](#) is an example of an appropriation bill. See Congress' [Appropriations and Budget Resources](#) for more information.

4.4.4 Appropriation Accounts

Federal departments and agencies are divided into organizational units such as bureaus, services, administrations, and offices. The federal budget is organized into appropriation and fund accounts that provide the link between legislation enacted by Congress and the financial operations of federal agencies. An appropriation account identifies a particular source of budget authority and the purposes for which that authority may be used. A single department or bureau may therefore operate through several separate accounts, each associated with particular programs, activities, or types of expenditure.

Appropriation accounts are important because Congress generally does not provide an agency with one unrestricted pool of money. Instead, Congress divides budget authority among accounts and specifies the purposes, amounts, and periods for which the funds are available. Agencies must obligate and spend funds consistently with those statutory restrictions. The exact organizational relationship differs across agencies, but a useful simplified representation is a flow starting with the department (e.g., Department of Energy) followed by the bureau or agency and then going into an appropriation account to be spent on a program or activity. The Department of the Interior, for example, contains several bureaus, and each bureau may receive budget authority through multiple accounts.

4.4.5 Budget Baselines

A budget baseline is a benchmark projection of federal revenues, spending, deficits, and debt under assumptions specified in budget law. It is not intended to be a prediction of what policymakers will actually do. Instead, it provides a reference point against which proposed legislation and alternative fiscal policies can be evaluated. The CBO constructs its baseline using rules established in federal law, including assumptions about the continuation of many mandatory programs and the future path of discretionary funding.

The idea of a current services baseline is closely related but should be distinguished from a simple forecast. A current-services estimate asks, in broad terms, what resources would be required to continue existing governmental services or policies under specified assumptions about prices, workloads, participation, and other factors. Because baseline construction is governed by particular statutory conventions, a formal CBO baseline can differ from an analyst's estimate of what would be required literally to maintain current services. For budget analysis, the baseline is therefore best understood as a standardized policy benchmark rather than as the most likely future budget outcome. Although more intended for states, the following publication by the Center on Budget and Policy Priorities provides an overview of [The Current Services Baseline: A Tool for Understanding Budget Choices](#)). It serves as a baseline or benchmark *reflecting this year's enacted policies and programs for each year in the future* and are part of the supporting material for budgets.

4.5 Budget Authority

Budget authority is the authority provided by federal law that allows the government to enter into financial obligations that will result in immediate or future outlays. Budget authority should be distinguished from outlays. Budget authority permits an agency to incur an obligation, whereas an outlay occurs when the federal government actually makes a payment to satisfy an obligation. Consequently, budget authority provided in one fiscal year may produce outlays in that year or in later years.

Appropriations are the most familiar form of budget authority. An appropriation generally authorizes an agency to incur obligations and make payments from the Treasury for specified purposes. Federal law also provides other forms of budget authority. *Borrowing authority* permits a federal entity to borrow funds and use the proceeds to incur and liquidate obligations. *Contract authority* permits an agency to enter into obligations before Congress provides the cash needed to make the resulting payments; subsequent appropriations or collections are then used to liquidate those obligations. *Spending authority* from offsetting collections permits an agency to incur obligations and make payments using certain collections credited to an expenditure account. These distinctions matter because not all federal programs receive their budget authority through annual

appropriation acts.

Appropriations differ in the length of time during which agencies may incur new obligations. A one-year, or annual, appropriation is available for obligation only during a specific fiscal year. A multi-year appropriation remains available for a fixed period longer than one fiscal year, such as two or three years. A no-year appropriation remains available for an indefinite period and commonly contains language such as “to remain available until expended.” The period of availability concerns when an agency may create new obligations; it is distinct from the period during which previously incurred obligations may be paid. Once the period of availability for a fixed-period appropriation ends, the appropriation becomes expired and is no longer available for new obligations. It generally remains available for a limited period to record, adjust, and liquidate obligations that were properly incurred before expiration. This distinction between available, expired, and closed accounts is part of the system used to prevent agencies from treating appropriated funds as permanently available resources.

The timing of an appropriation is a separate issue from its period of availability. An [advance appropriation](#) is enacted in one fiscal year but does not become available for obligation until a later fiscal year. Congress can use advance appropriations when it wants to commit future budget authority while delaying the date on which agencies may use it. A permanent appropriation is budget authority that becomes available under existing law without Congress having to enact a new appropriation for that purpose each year. Permanent appropriations are commonly associated with programs or payments for which the underlying law provides continuing financing. A permanent appropriation can still be subject to statutory rules determining the amount that becomes available. Thus, the term permanent describes the continuing legal availability of the authority rather than implying that Congress can never change it.

The federal fiscal year begins on October 1. If one or more regular appropriation acts have not been enacted by that date, Congress may enact a continuing resolution, commonly called a CR, to provide temporary budget authority. A continuing resolution prevents affected agencies and programs from immediately losing funding while Congress completes action on regular appropriations. Continuing resolutions typically provide funding for a limited period and commonly base funding on specified rates or formulas related to previous appropriations. They often restrict or complicate the initiation of new programs, projects, or activities. A short continuing resolution may have relatively modest operational consequences, whereas repeated or lengthy continuing resolutions can make planning, contracting, hiring, and program management more difficult because agencies do not know their final full-year funding levels.

Congress may also provide supplemental appropriations after regular appropriations have been enacted. Supplemental appropriations provide additional budget authority for needs that were not fully anticipated or funded in the regular appropriation process. They are frequently associated with emergencies or unusually large and uncertain expenditures, including responses to natural



Figure 4.1: Share of mandatory spending over time.

disasters, military operations, economic crises, and public-health emergencies. Supplemental appropriations can also be used when the cost of an existing activity turns out to be greater than initially expected.

4.6 Discretionary and Mandatory Spending

Federal outlays are commonly divided into three broad categories: discretionary spending, mandatory spending, and net interest. This classification concerns how spending is controlled in the budget process rather than what the government purchases. Discretionary spending is controlled primarily through appropriation acts. Congress therefore determines funding for most discretionary programs by specifying amounts in appropriations legislation. Defense, many federal agency operations, scientific research, environmental programs, education grants, and numerous transportation and infrastructure activities contain substantial discretionary components.

Mandatory spending, also called direct spending, is generally governed by statutory criteria rather than by specific amounts determined through the annual appropriation process. For many mandatory programs, Congress establishes eligibility requirements, benefit formulas, payment rules, or other parameters in authorizing legislation. Actual spending then depends on the number of eligible participants, economic conditions, benefit levels, and other factors. Social Security, Medicare, Medicaid, unemployment insurance, and several income-support programs are major examples. Congress can change mandatory spending, but

it normally does so by amending the laws that determine eligibility, benefits, payments, or program financing rather than by simply changing an annual appropriation amount.

The distinction between mandatory and discretionary spending is therefore not the same as the distinction between important and unimportant spending, nor does mandatory mean that the spending can never be changed. Both categories are created and can be altered by law. The difference is the mechanism through which spending is determined. Discretionary programs are generally funded through appropriation acts, whereas mandatory programs generally generate spending according to provisions of laws that remain in effect until Congress changes them.

The composition of federal spending has changed substantially over time. Mandatory programs have grown as a share of total federal outlays, reflecting the expansion and maturation of federal retirement and health programs, demographic change, and rising health expenditures. Because mandatory spending is generally driven by statutory eligibility and benefit rules, much of this spending occurs without the need for Congress to set a new dollar amount in the annual appropriations process. This helps explain why the share of federal spending controlled directly through annual appropriations is smaller today than it was in earlier periods. For outlooks about the future of spending, the CBO provides an [Outlook for the Budget and the Economy](#) on a regular basis.

4.7 Major Mandatory Social Insurance and Health Programs

Social Security, Medicare, and Medicaid account for a large portion of federal mandatory spending, but they differ substantially in their purposes, eligibility rules, and financing. Social Security is primarily a social insurance program providing cash benefits related to covered earnings and work history. Medicare is a federal health insurance program with different financing arrangements across its components. Medicaid is a means-tested health program jointly financed by the federal government and the states.

Mandatory benefit programs can also be classified according to how eligibility is determined. In a means-tested program, eligibility or the amount of assistance depends partly on income, assets, or another measure of economic need. Medicaid and the Supplemental Nutrition Assistance Program are examples of programs in which economic circumstances play an important role in eligibility. In a non-means-tested social insurance program, eligibility is based primarily on criteria such as age, disability, covered employment, or work history rather than on current income or wealth. Social Security and Medicare are important examples, although particular components or subsidies within large programs may themselves be income related.

4.7.1 Social Security (OASDI)

The Old-Age, Survivors, and Disability Insurance program, or OASDI, is generally known as Social Security. It consists of two legally separate trust funds: the Old-Age and Survivors Insurance Trust Fund and the Disability Insurance Trust Fund. OASI pays benefits to retired workers, eligible family members, and survivors of deceased workers. DI pays benefits to workers who meet the program's insured-status and disability requirements and to certain eligible family members.

Social Security is financed primarily through payroll taxes levied on covered earnings, together with revenue from the taxation of some Social Security benefits and interest credited to the trust funds. The relationship between annual income and annual benefit payments therefore depends heavily on employment, taxable earnings, demographic conditions, and the number of beneficiaries. When current noninterest income is insufficient to cover scheduled benefits and administrative costs, trust fund assets can be redeemed to cover the difference as long as reserves remain available. See the Social Security Administration's [Annual Report](#).

4.7.2 Medicare

Medicare is a federal health insurance program primarily serving people age 65 and older, as well as certain younger people who meet disability or other statutory eligibility requirements. Medicare has several components. Part A, Hospital Insurance, primarily covers inpatient hospital services and certain related services. Part B, Supplementary Medical Insurance, covers physician services, outpatient care, and other medical services. Part C, Medicare Advantage, allows beneficiaries to receive Medicare-covered benefits through participating private plans. Part D provides outpatient prescription-drug coverage.

Medicare does not have one uniform financing mechanism. Part A is financed primarily through payroll taxes and is associated with the Hospital Insurance Trust Fund. Parts B and D are financed mainly through beneficiary premiums and transfers from the federal government's general fund and are associated with the Supplementary Medical Insurance Trust Fund. Because general revenues finance a substantial portion of Parts B and D, Medicare's financing structure differs importantly from Social Security even though both programs are classified largely as mandatory spending.

4.7.3 Medicaid

Medicaid is a joint federal-state program providing health coverage to eligible low-income individuals and to other groups that qualify under federal and state rules. Unlike Medicare, Medicaid is not primarily an age-based social insurance program and does not rely on a dedicated federal payroll tax or federal trust fund. Federal and state governments share program costs, with the federal government reimbursing states for a specified share of eligible expenditures.

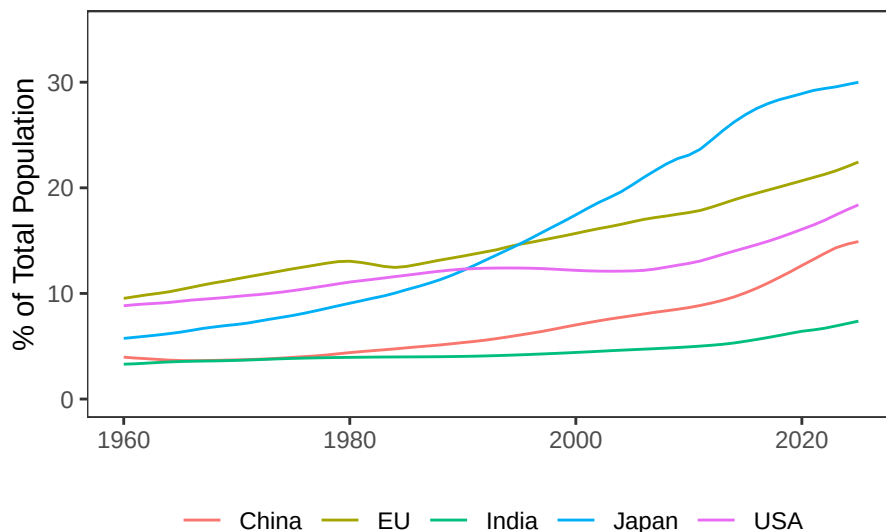


Figure 4.2: Population Age 65 and over

The federal matching share varies according to statutory formulas and can differ across populations and services. States administer Medicaid within broad federal requirements and therefore have some discretion over program design, eligibility beyond federal minimum requirements, provider payments, and covered services. Federal Medicaid spending is mandatory because federal payments to states generally follow the eligibility, matching, and payment rules established in law rather than being determined by a single annual discretionary appropriation.

4.8 Financing Challenges for Social Insurance and Health Programs

Social Security is often described as largely pay-as-you-go financed because payroll taxes and other current income are used to finance current benefits, while trust fund reserves provide a mechanism for carrying surpluses or deficits across time. The system is therefore not a pure pay-as-you-go arrangement: trust fund assets matter when annual program income differs from annual cost. A central long-run issue is the relationship between the number of workers paying payroll taxes and the number of beneficiaries receiving benefits. Population aging, low fertility, mortality trends, immigration, labor-force participation, and wage growth all influence that relationship.

Medicare and Medicaid face a somewhat different set of financing pressures because their expenditures depend not only on the number of eligible benefi-

ciaries but also on the cost and utilization of health-care services. Population aging increases enrollment in Medicare, while growth in health-care costs per beneficiary can raise spending even if the number of beneficiaries is unchanged. Medicaid spending is influenced by enrollment, economic conditions, health-care costs, federal-state matching rules, and policy choices made by both levels of government. Consequently, demographic change and health-cost growth are major determinants of the long-run federal budget outlook. The CBO published a report outlining [Mandatory Spending Control Mechanisms](#).

4.8.1 Pay-As-You-Go Financing and Prefunding

Pay-as-you-go financing and prefunding are useful conceptual models for thinking about retirement systems. Under a pure pay-as-you-go system, contributions collected from current workers are used to finance benefits paid to current retirees. Under a fully prefunded system, contributions are accumulated and invested in advance so that assets are built up to finance future benefits. Actual public programs can combine characteristics of both approaches.

Social Security is largely financed from current payroll-tax and other program income, but it also maintains trust funds that can accumulate reserves when income exceeds cost and redeem reserves when cost exceeds current income. It should therefore not be described as either a purely pay-as-you-go system or a fully funded individual retirement system. The distinction is nevertheless useful because demographic change has particularly direct consequences for systems that rely heavily on transfers from current workers to current beneficiaries.

Prefunding is easiest to apply conceptually to retirement benefits because contributions and expected benefits can be related over a worker's life cycle. The concept is less directly applicable to Medicaid because Medicaid is a means-tested health program rather than a contributory retirement program. Medicare also cannot be characterized by one simple funding model because Part A relies heavily on payroll-tax financing and a trust fund, whereas Parts B and D rely substantially on premiums and general revenues.

4.9 Federal Debt

A federal budget deficit occurs when federal outlays exceed federal revenues during a fiscal year. The Treasury generally finances deficits by borrowing, which increases federal debt. The debt has recently surpassed \$40 trillion and the interest payments are rising as well. A surplus has the opposite effect, although changes in debt can also reflect other financial transactions. For economic analysis, an especially important measure is debt held by the public, which consists largely of Treasury securities held by investors outside the federal government. This measure differs from gross federal debt, which also includes Treasury securities held by federal government accounts such as trust funds.

Federal debt affects the budget directly through interest payments. As the stock

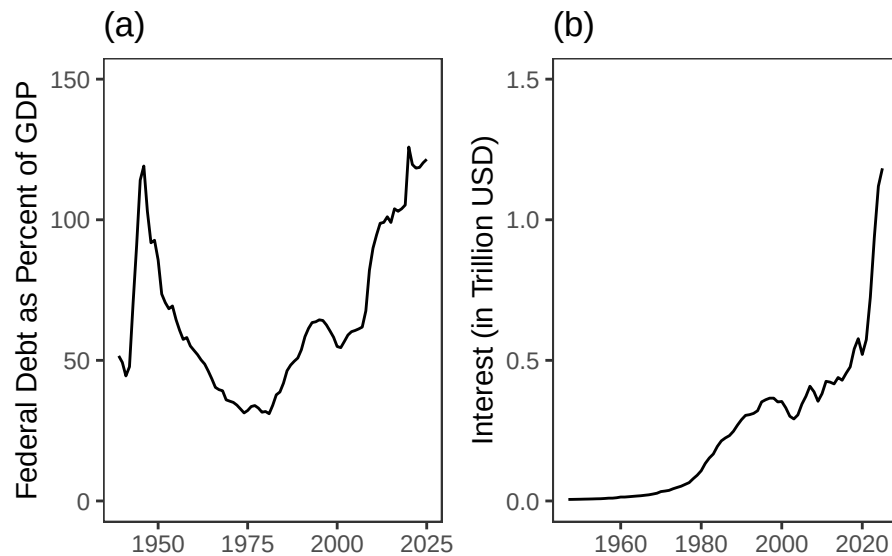


Figure 4.3: Panel (a): Share of mandatory spending over time. Panel (b): Federal government current expenditures: Interest payments.

of debt or the interest rates paid on that debt increase, federal net interest outlays tend to rise. Higher interest costs can in turn contribute to larger deficits and additional borrowing. This creates an important connection between past borrowing decisions, current interest rates, and the amount of future budgetary resources available for other federal activities.

Federal borrowing can also affect the broader economy. When government borrowing increases the overall demand for funds in financial markets, it can put upward pressure on interest rates and reduce, or crowd out, some private investment. Less private capital formation can reduce future productive capacity and income relative to what they otherwise would have been. The size of these effects depends on economic conditions, monetary policy, international capital flows, and how the borrowed funds are used.

A high level of debt can also reduce fiscal flexibility. When the government enters a recession, war, financial crisis, public-health emergency, or other major event with already elevated debt and interest costs, policymakers may face more difficult choices about additional borrowing, taxes, and spending. Debt should therefore be evaluated not only by its dollar value but also relative to the size of the economy, the government's revenue base, interest rates, and the expected future path of spending and revenues.

4.10 Exercises

1. **Agency Budget III** (**): Use one of the following federal agencies for this exercise: Fish and Wildlife Service, Forest Service, Bureau of Land Management, Bureau of Reclamation, National Oceanic and Atmospheric Administration, U.S. Geological Survey, Army Corps of Engineers, Environmental Protection Agency, and U.S. Coast Guard. You will be able to locate the answers in the Budget of the United States Government, available through the OMB website.
 - a. Determine where each agency lies in the administrative structure of the federal government. Is it in an executive department or is it an independent agency?
 - b. Determine which of the appropriation subcommittees has jurisdiction over the agency's budget request.
 - c. Determine where the agency's operations fall in the functional classification of the budget.
 - d. For a recent budget year, determine (1) the budget authority and outlays proposed for the agency, and (2) the actual outlays and budget authority for the agency in that year.
2. **Federal Budget Deficit** (**): Read the August 2018 report from the Congressional Budget Office (CBO) which is titled [The Deficit Reductions Necessary to Meet Various Targets for Federal Debt](#). Note that the table of content indicates a "Summary" which contains most of the answers to the following questions:
 - a. What is the difference between constant-share scenarios and growing-share scenarios?
 - b. What are the effects on income per person (Gross National Product or GNP per person in the report) from reducing the deficit under various scenarios?
 - c. Reducing the federal deficit can be accomplished by either increasing taxes or decreasing spending. What do you think are some of the economic, demographic, and political difficulties that arise while deciding which deficit-reducing policy to implement?
 - d. How has the COVID-19 pandemic and its associated change in the economic and fiscal environment changed the projections outlined in the report.
3. **State Budgets** (**): Pick a state of your choice (not Indiana) and identify key elements (listed below) of your state's budget process and budget. The National Association of State Budget Officers (NASBO) provides much of this information in its publication [Budget Processes in the States](#).
 - a. Does your state have an annual or biennial budget?
 - b. Which functions does the budget agency execute in your state? E.g, economic analysis, audits?

- c. What veto power, if any, does the governor have? How many votes does it take to override the governor's veto?
 - d. Is the state's budget easily accessible on a state run website?
 - e. Access the most recent (proposed) budget document and identify, to the extent possible, what policy priorities are identified in the report.
4. **Budget Authority and Outlays** (***) : Select one federal agency or major program and obtain budget authority and outlay data for at least five consecutive fiscal years from an official federal source. For each year, calculate the ratio of outlays to budget authority. Explain why the two quantities need not be equal in a given fiscal year. Identify any year in which the difference is especially large and investigate the reason for the difference. Your explanation should consider the timing of obligations and outlays, the availability period of appropriations, and whether the program contains one-year, multiyear, or no-year budget authority. Next, provide your data and calculations to an AI system and ask it to explain the observed differences between budget authority and outlays. Evaluate the AI response. Identify at least one statement that is well supported, one statement that is incomplete or overly general, and any statement that cannot be verified from the official budget documents you used. <https://www.govinfo.gov/app/details/BUDGET-2027-APP>
5. **Constructing a Federal Budget Baseline** (***) : Suppose federal revenues in year 0 are \$5.0 trillion and federal outlays are \$6.5 trillion. Of total outlays, \$1.8 trillion is discretionary spending, \$3.8 trillion is mandatory spending, and \$0.9 trillion is net interest. Construct a 10-year baseline using the following assumptions. Revenues grow by 4% per year. Mandatory spending grows by 5% per year. Discretionary spending grows by 2% per year. The average interest rate on federal debt is 3.5% in year 1 and increases by 0.25 percentage points each subsequent year. Initial debt held by the public is \$40 trillion. For each year, calculate revenues, discretionary spending, mandatory spending, net interest, total outlays, the budget deficit, and debt held by the public. Assume, for purposes of this exercise, that the entire annual deficit is added to debt held by the public and that interest in each year is calculated using debt at the beginning of that year. Then modify the baseline by imposing a policy that reduces annual discretionary spending growth from 2% to zero beginning in year 2. Calculate the cumulative reduction in deficits and the difference in debt at the end of year 10. After completing the exercise yourself, ask an AI system to perform the same projection. Compare its results with yours. Do not simply report whether the answers differ. Locate the first year and variable in which any discrepancy arises and determine whether the error results from arithmetic, compounding, timing of interest payments, or interpretation of the assumptions. Note that the purpose of this exercise is to demonstrate the effects that small changes in percentage values can have over time and the magnitude of the budget in very simplified terms.

6. ***Mandatory Spending and Demographic Change*** (***) : Consider a simplified mandatory retirement program. In year 0, there are 170 million covered workers and 70 million beneficiaries. The average taxable payroll per worker is \$60,000, the payroll tax rate supporting the program is 12 percent, and the average annual benefit is \$24,000.
- Calculate total payroll-tax revenue, total benefit expenditures, and the program's annual balance.
 - Now project the program ten years forward under the following assumptions. The number of workers grows by 0.3 percent per year, the number of beneficiaries grows by 2.0 percent per year, taxable payroll per worker grows by 3.0 percent per year, and the average benefit grows by 3.0 percent per year. Hold the payroll tax rate constant.
 - Calculate the worker-to-beneficiary ratio and the program balance in year 10. Explain why the financial position of a largely pay-as-you-go system can deteriorate even when wages and payroll-tax revenues are increasing.
 - Next, determine the payroll tax rate that would be required in year 10 to make payroll-tax revenues exactly equal to benefit expenditures under these assumptions.
 - Ask an AI system to analyze the same scenario. Your submission must include a short audit of its answer. In particular, determine whether the AI correctly compounds population and wage growth, distinguishes the growth rate of the number of workers from the growth rate of payroll per worker, and calculates the balancing payroll tax rate from year-10 values rather than from year-0 values.